

Lining Epithelium

I- Introduction:

A lining epithelium is formed of joined, juxtaposed cells, united to each other by junction systems and separated from the underlying connective tissue by a basement membrane. It can:

- Either isolate other tissues from the external environment: this is the case of the epidermis.
- Or line the natural cavities of the organism such as: the oral cavity, esophagus, stomach...

II - Embryological Origin

The tissues of the organism develop from the three primitive embryonic layers:

- ✦ The ectoblast provides the skin and the nervous system.
- ✦ The endoblast provides the digestive tract and the pulmonary system.
- ✦ The mesoblast provides the muscles, skeleton, cartilage, and the urogenital system.

III- Morphology of Lining Epithelia:

A- General classification criteria: Three criteria allow classification: Cell shape + Number of cell layers + Specialization of the apical pole

1- Cell shape:

- Squamous: wider than tall.
- Cuboidal: as tall as wide.
- Columnar (Cylindrique) or prismatic: taller than wide.

2- Number of cell layers:

- Simple epithelium: single cell layer.
- Stratified epithelium: several layers; named according to the most superficial cell type.

+ Thus we have:

- Stratified squamous epithelia
- Stratified cuboidal epithelia
- Stratified columnar epithelia

3- Apical pole specializations: Microvilli, motile cilia, stereocilia, mucous cells.

Microvilli:

Short projections of the plasma membrane covering cytoplasmic extensions.

They may be isolated or grouped forming:

- Striated border (intestinal epithelium)
- Brush border (renal proximal tubules)

Motile cilia:

Mobile cytoplasmic extensions with pendular or undulating movement.

They move particles and fluids. Example: respiratory tract, testicular excretory ducts.

Stereocilia:

Long, immobile cytoplasmic projections forming tufts.

Mucous cells: Two types:

- Open mucous pole cells (goblet cells)
- Closed mucous pole cells

B- Analytical Study :

1- Squamous epithelia:

a- Simple squamous epithelium:

Composed of flat cells wider than they are tall, arranged in a single layer. They line the parietal and visceral layers of serous membranes (pleura, pericardium, peritoneum), known as the mesothelium. They also line vascular cavities, known as endothelia

b- Stratified squamous epithelium (Malpighian):

○ **Keratinized type:**

Epidermis of the skin.

From the basement membrane to the surface, the layers are:

- Germinative layer (cuboidal cells)
- Spinous layer = prickle cell layer (polygonal cells with intercellular bridges)
- Granular layer (flattened cells with granules)
- Clear layer
- Cornified layer = corneum layer (dead flattened cells)

○ **Non-keratinized type:**

Lines oral cavity, esophagus, vagina, cornea. It consists of four layers:

- ✦ basal (CB)
- ✦ Parabasal (CPB)
- ✦ Intermediate (CI)
- ✦ Superficial (CS)

2- Cuboidal epithelia:

- Simple cuboidal epithelium: One layer of cells as tall as they are wide with round nuclei. Example: ovarian germinal epithelium.
- Stratified cuboidal epithelium: Rare, generally featuring two layers, with the superficial one being cuboidal. Example: sweat gland ducts.

3- Columnar (Prismatic) Epithelia:

- Simple Columnar Epithelia: One layer of cells taller than they are wide with ovoid nuclei. Examples:
 - ✦ Bile ducts
 - ✦ The gastric epithelium (closed mucous pole)
 - ✦ The intestinal epithelium (striated border and goblet cells)
 - ✦ The fallopian tube (ciliated).
- Stratified Columnar Epithelia: Several layers with a superficial columnar layer. Example: male urethra.

4- Pseudostratified epithelia:

Appear stratified but all cells rest on the basement membrane:

- Ciliated pseudostratified columnar epithelium: respiratory tract
- Urothelium (Transitional Epithelium): Appears multi-layered; all cells attach to the basal lamina via a slender expansion. It can change shape by cells sliding over one another (bladder, ureters)
- Pseudostratified epithelium with stereocilia: epididymis

IV- Properties:

1. Renewal: Cells have a short lifespan and are continuously renewed by mitosis of cells located against the basal lamina.

2. Permeability: Epithelia are avascular; nutrition is only possible through the diffusion of substances from blood capillaries in the underlying connective tissue.

3. Cohesion: Functional differentiations on lateral faces increase cohesion, close intercellular spaces, or facilitate exchanges.

- ✦ Interdigitations: Plasma membranes of adjacent cells interlock.
- ✦ Cell Junctions:
 - Group 1 (Fusion): Includes tight junctions (zonula occludens) which can be pentalamellar and punctiform.
 - Group 2 (No Fusion): Includes intermediate junctions (zonula adhaerens) and desmosomes (macula adhaerens). Hemidesmosomes attach the epithelium to the basal lamina.
- ✦ Junction Complexes: Association of three types (e.g., in enterocytes: zonula occludens, zonula adhaerens, and macula adhaerens)

V- Functions:

1. Protection: Mechanical/thermal (epidermis), chemical (gastric epithelium), and physical (melanin against solar radiation).
2. Absorption: (e.g., intestinal epithelium).
3. Sensory Reception: For touch, temperature, pain (epidermis), smell (olfactory epithelium), and taste (taste buds).
4. Secretion: Creating new products from blood elements and excreting them (e.g., intestinal epithelium).
5. Excretion: Eliminating toxic waste from the blood (e.g., renal epithelium).
6. Exchange: Between internal and external environments.
7. Movement: Using vibratile cilia to move mucus or dust (e.g., respiratory epithelium).
8. Sliding: Mesothelia (pleural, pericardial, peritoneal) secrete fluid to allow tissues to slide over each other.

VI- Epithelial Transformations:

1. Hyperplasia: Excessive proliferation leading to increased thickness (e.g., hormonal stimulation in the uterus or mammary gland).
2. Metaplasia: Transformation of one epithelium type into another (e.g., in smokers, respiratory epithelium may transform into non-keratinized stratified squamous epithelium)